

Table 1: ARI on dataset collection 2 with 35.0% constraints

| Constraints Level<br>Method<br>Dataset | 0.35    |      |        |           |             |         |      |       |
|----------------------------------------|---------|------|--------|-----------|-------------|---------|------|-------|
|                                        | COPKM_P | KSKM | KSKM_E | KSKM_repo | KSKM_E_repo | BLPKMCC | PCCC | COPKM |
| synthetic_n300_d2_k10_s10              | 1.00    | 1.00 | 1.00   | 1.00      | 1.00        | 1.00    | 1.00 | 1.00  |
| synthetic_n300_d2_k10_s20              | 1.00    | 1.00 | 1.00   | 1.00      | 1.00        | 1.00    | 1.00 | 1.00  |
| synthetic_n300_d2_k10_s30              | 1.00    | 1.00 | 1.00   | 1.00      | 1.00        | 1.00    | 1.00 | 1.00  |
| synthetic_n300_d2_k10_s40              | 1.00    | 1.00 | 1.00   | 1.00      | 1.00        | 1.00    | 1.00 | 1.00  |
| synthetic_n300_d2_k10_s50              | 1.00    | 1.00 | 1.00   | 1.00      | 1.00        | 1.00    | 1.00 | 1.00  |
| synthetic_n300_d2_k20_s10              | 1.00    | 1.00 | 1.00   | 1.00      | 1.00        | 1.00    | 1.00 | 1.00  |
| synthetic_n300_d2_k20_s20              | 0.99    | 0.99 | 0.99   | 0.99      | 0.99        | 0.99    | 0.99 | 0.99  |
| synthetic_n300_d2_k20_s30              | 0.93    | 0.93 | 0.92   | 0.93      | 0.93        | 0.93    | 0.93 | 0.93  |
| synthetic_n300_d2_k20_s40              | 0.89    | 0.90 | 0.90   | 0.90      | 0.90        | 0.90    | 0.89 | -     |
| synthetic_n300_d2_k20_s50              | 0.87    | 0.87 | 0.87   | 0.86      | 0.86        | 0.87    | 0.87 | -     |
| synthetic_n300_d2_k50_s10              | 0.81    | 0.89 | 0.86   | 0.87      | 0.87        | 0.78    | 0.85 | 0.81  |
| synthetic_n300_d2_k50_s20              | 0.63    | 0.67 | 0.69   | 0.66      | 0.66        | 0.61    | 0.62 | 0.59  |
| synthetic_n300_d2_k50_s30              | 0.47    | 0.47 | 0.46   | 0.47      | 0.46        | 0.47    | 0.47 | 0.44  |
| synthetic_n300_d2_k50_s40              | 0.38    | 0.40 | 0.40   | 0.40      | 0.39        | 0.39    | 0.38 | 0.38  |
| synthetic_n300_d2_k50_s50              | 0.36    | 0.36 | 0.36   | 0.36      | 0.35        | 0.36    | 0.35 | 0.35  |